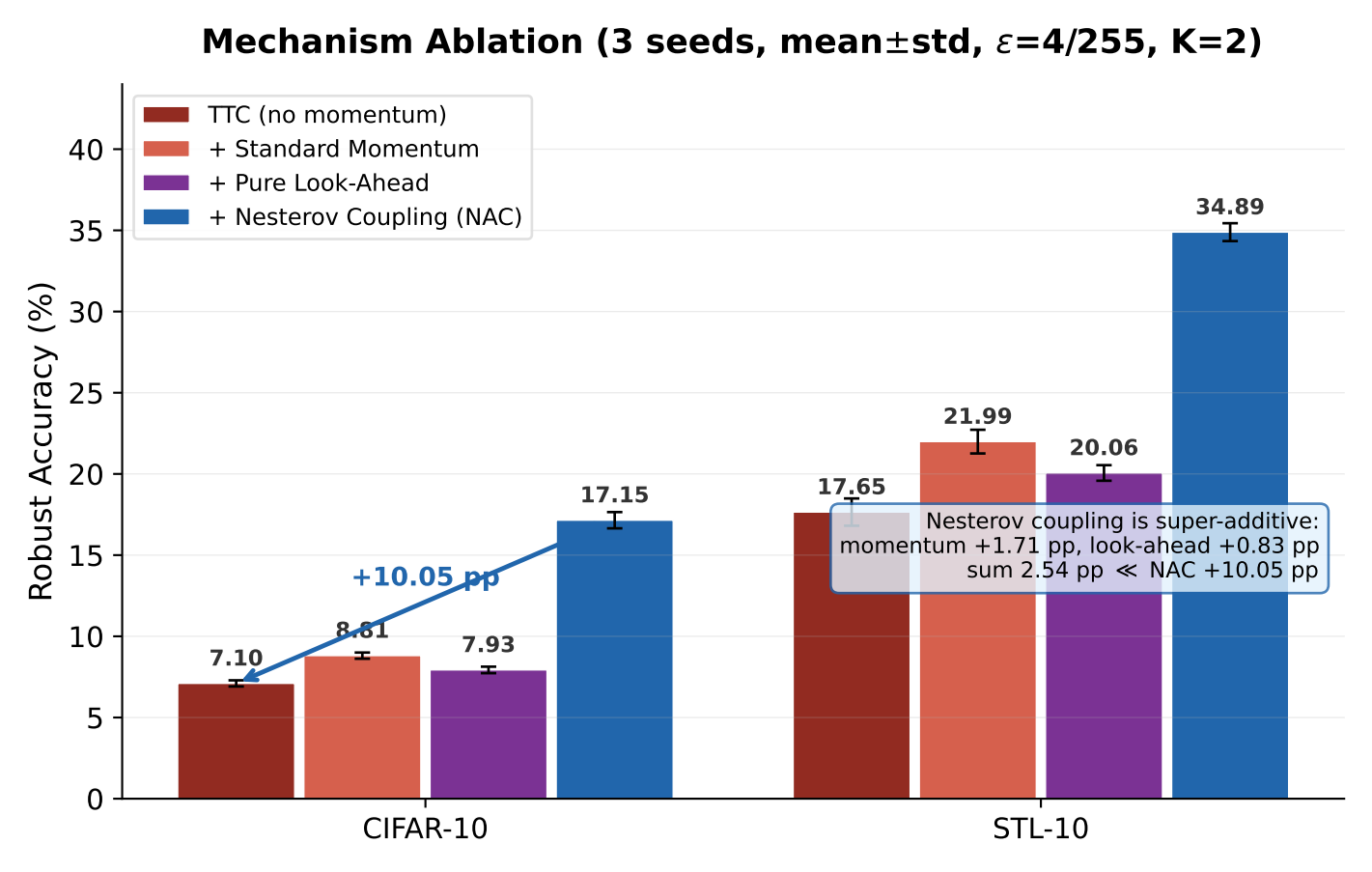
Mechanism Ablation (3 seeds, mean±std, ε=4/255, K=2)
TTC (no momentum)
+ Standard Momentum
+ Pure Look-Ahead
+ Nesterov Coupling (NAC)
Robust Accuracy (%)
0
5
10
15
20
25
30
35
40
7.10
8.81
7.93
17.15
+10.05 pp
17.65
21.99
20.06
34.89
Nesterov coupling is super-additive:
momentum +1.71 pp, look-ahead +0.83 pp
sum 2.54 pp ≪ NAC +10.05 pp
CIFAR-10
STL-10